

DATABASE SCHEMA DOCUMENTATION

Production-Ready Schema Design

Version 1.0 · June 2026

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1. System Overview

This document defines the complete relational database schema for the business management platform. The system supports five operational domains: Marketing, Finance, CEO Administration, Design, and Quantity Surveying. It models the full lifecycle of customer acquisition, payment tracking, task assignment, submission, and review across all departments.

1.1 Database Technology Assumptions

- Primary Database: PostgreSQL 15+
- UUID generation: `gen_random_uuid()` (pgcrypto extension)
- Array columns: Native PostgreSQL `TEXT[]` for file attachment URL arrays
- Timezone: All timestamps stored as `TIMESTAMPTZ` (UTC), displayed in EAT/UTC+3
- Character Encoding: UTF-8

1.2 File Attachment Strategy

The document specifies that all file attachments are stored as arrays of strings. In this schema, this is implemented using the PostgreSQL `TEXT[]` (text array) column type. Each element in the array is a fully-qualified URL pointing to a file stored in an external object storage service (e.g., AWS S3, Azure Blob Storage, or Google Cloud Storage).

2. Table Schema Definitions

2.1 Core / Identity

Table: user

Purpose: Central user registry. Every actor in the system (marketing staff, finance, CEO, designers, quantity surveyors, data collectors, reviewers) is represented here. Role-based access is derived from the role column.

Column Name	Data Type	Key	Nullable	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique user identifier
full_name	VARCHAR(255)		NO	NOT NULL	Display name of the user
email	VARCHAR(255)		NO	UNIQUE, NOT NULL	Login email; must be unique system-wide
phone	VARCHAR(20)		YES	E.164 format	Optional contact phone
role	ENUM('marketing','finance','ceo','designer','quantity_surveyor','data_collector','general_manager')		NO	NOT NULL	System role governing access permissions
is_active	BOOLEAN		NO	DEFAULT TRUE	Soft-disable without deleting the record
password_hash	VARCHAR(255)		NO	NOT NULL	Bcrypt/Argon2 hashed password
last_login_at	TIMESTAMPTZ		YES		Timestamp of most recent successful login
created_at	TIMESTAMPTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPTZ		Yes	DEFAULT NOW()	Timestamp of last update (UTC)
created_by	UUID	FK	NO	FK → user.id	User who created this record

Indexes:

```
CREATE INDEX idx_user_name_role ON "user" (role);
```

2.2 Marketing

Table: customer

Purpose: Stores prospective and active customer enquiries. Each row represents one customer contact/lead captured by the marketing team.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique customer record identifier
marketing_user_id	UUID	FK	NO	FK → user.id (role=marketing)	Marketing staff member who captured this lead
customer_name	VARCHAR(500)		NO	1–500 chars, trimmed	Full name or company name of the customer
customer_phone	VARCHAR(20)		NO	E.164 format	Primary contact phone in E.164 format
customer_email	VARCHAR(255)		YES	Valid email format	Contact email address (optional)
customer_address	VARCHAR(2000)		NO	1–2000 chars	Full physical address of the customer
category	String		NO	NOT NULL	Service category the customer is interested in
service_description	TEXT		NO	1–5000 chars	Detailed description of the services requested
preferred_start_date	DATE		YES	ISO 8601, must not be in the past	Customer's preferred project start date
budget	NUMERIC(15,2)		YES	Positive, max 2 decimals	Estimated budget in local currency
notes	TEXT		YES	Max 5000 chars	Additional notes about the customer or request
paid	boolean		no	Default false	
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

```
CREATE INDEX idx_customer_marketing_user_id ON customer (marketing_user_id);
```

Table: paid_customer

Purpose: Records confirmed payment intent or received payment linked to a customer enquiry. Tracks the lifecycle status of each payment from pending through to acceptance or rejection.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique paid customer record
customer_id	UUID	FK	NO	FK → customer.id	The originating customer enquiry
description	TEXT		NO	NOT NULL	Description of the payment or agreement
status	ENUM('approved','rejected','feedback','pending')		NO	DEFAULT pending	Current lifecycle status of the payment
attachment_urls	TEXT[]		YES	Array of valid URLs	File attachment URLs (payment proofs, contracts). Stored as PostgreSQL text array.
created_at	TIMESTAMP PTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMP PTZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

```
CREATE INDEX idx_paid_customer_customer_status ON paid_customer (customer_id, status);
```

Table: marketing_submission

Purpose: Captures each submission event made by the marketing team for a paid customer record. A new submission updates the parent paid_customer status. Supports iterative submission cycles (e.g., resubmit after feedback).

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique submission identifier

Column Name	Data Type	Key	Null able	Constraints	Description
paid_customer_id	UUID	FK	NO	FK → paid_customer.id	The paid customer record being submitted
description	TEXT		NO	NOT NULL	Details of what is being submitted in this cycle
attachment_urls	TEXT[]		YES	Array of valid URLs	Supporting documents for this submission round
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

Indexes:

```
CREATE INDEX idx_marketing_submission_paid_customer_created_at
ON marketing_submission (paid_customer_id, created_at);
```

Table: marketing_review

Purpose: Stores reviewer decisions on marketing submissions. Each review is tied to a paid_customer record and records the outcome (approved, rejected, or feedback required) along with the reviewer's identity.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique review record
marketing_submission_id	UUID	FK	NO	FK → paid_customer.id	The marketing submission record under review
reviewer_user_id	UUID	FK	NO	FK → user.id	The user performing the review
description	TEXT		NO	NOT NULL	Reviewer's comments and observations

Column Name	Data Type	Key	Null able	Constraints	Description
review_outcome	ENUM('approved','rejected','feedback','pending')		NO	NOT NULL	Final decision of this review cycle
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

CREATE INDEX idx_marketing_review_submission_id ON marketing_review (marketing_submission_id);

2.3 Finance

Table: ceo_transfer

Purpose: Records financial transfer instructions or approvals initiated by the Finance department and directed to the CEO for authorization. Attachments hold supporting financial documents.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique transfer record
finance_user_id	UUID	FK	NO	FK → user.id (role=finance)	Finance team member initiating the transfer
ceo_user_id	UUID	FK	NO	FK → user.id (role=ceo)	CEO user receiving the transfer request
description	TEXT		NO	NOT NULL	Description of the financial transfer or its purpose
amount	NUMERIC(15,2)		No	Positive	Transfer amount (assumed based on finance context)
attachment_urls	TEXT[]		YES	Array of valid URLs	Supporting financial documents as URL array

Column Name	Data Type	Key	Null able	Constraints	Description
created_at	TIMESTAMP PTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMP PTZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

2.4 CEO / Data Collector

Table: data_collector_task

Purpose: Tasks assigned by the CEO to data collectors. Represents a unit of work with defined scope, deadline, and expected deliverables.

Column Name	Data Type	Key	Nullable	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique task identifier
assigned_to_user_id	UUID	FK	yes	FK → user.id (role=data_collector)	Data collector assigned to this task
assigned_by_user_id	UUID	FK	NO	FK → user.id (role=ceo)	CEO who assigned the task
title	VARCHAR(500)		NO	NOT NULL	Short descriptive title of the task
description	TEXT		NO	NOT NULL	Detailed task instructions
status	ENUM(approve, feedback, rejected, pending)		yes	DEFAULT open	Current progress status of the task
task_state	Enum (active / deactive)		no	Default active	The current task is not rejected it is deactivated because of many reasons
due_date	DATE		YES		Target completion date
attachment_urls	TEXT[]		YES	Array of valid URLs	Reference documents attached to the task
created_at	TIMESTAMPTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPTZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

Column Name	Data Type	Key	Nullable	Constraints	Description
updated_by	UUID	FK	YES	FK → user.id	User who last modified this record

Indexes:

```
CREATE INDEX idx_data_collector_task_assigned_status ON data_collector_task (assigned_to_user_id, status);
```

Table: data_collector_submission

Purpose: Submissions made by data collectors against an assigned task. A task may have multiple submission cycles if feedback is returned.

Column Name	Data Type	Key	Nullable	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique submission identifier
data_collector_task_id	UUID	FK	NO	FK → data_collector_task.id	The task this submission belongs to
description	TEXT		NO	NOT NULL	Summary of work completed in this submission
attachment_urls	TEXT[]		YES	Array of valid URLs	Submitted data files or reports
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

```
CREATE INDEX idx_data_collector_submission_task_id ON data_collector_submission (data_collector_task_id);
```

Table: data_collector_review

Purpose: CEO review of a data collector submission. Records the approval decision and feedback for quality control.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique review record
data_collector_submission_id	UUID	FK	NO	FK → data_collector_submission.id	The submission being reviewed
reviewer_user_id	UUID	FK	NO	FK → user.id (role=ceo)	user performing the review
description	TEXT		NO	NOT NULL	Review comments and guidance
review_outcome	ENUM('approved','rejected','feedback','pending')		NO	NOT NULL	Decision on the submission
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

```
CREATE INDEX idx_data_collector_review_submission_id
ON data_collector_review (data_collector_submission_id);
```

```
CREATE INDEX idx_data_collector_review_reviewer_user_id
ON data_collector_review (reviewer_user_id);
```

2.5 CEO / Designer

Table: designer_task

Purpose: Design tasks assigned by the CEO to designer users. Tracks task lifecycle from assignment through completion.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique designer task identifier
assigned_to_user_id	UUID	FK	yes	FK → user.id (role=designer)	Designer assigned to this task

Column Name	Data Type	Key	Nulla ble	Constraints	Description
assigned_by_user_id	UUID	FK	NO	FK → user.id (role=ceo,general manger)	CEO who assigned the task
title	VARCHAR(500)		NO	NOT NULL	Short task title
description	TEXT		NO	NOT NULL	Detailed design brief
status	ENUM('approved','rejected','feedback','pending')		Yes	DEFAULT open	Current task status
stage	enum(case study, designing, rendering, final stage)		YES		The current stage that design task is
is_paused	boolean		yes		If the current task is active or paused
is_public	boolean		yes		If the current task is available for all users
task_state	Enum (active / deactive)				The current task is not rejected it is deactivated because of many reasons
Story point	int		no		
due_date	DATE		YES		Expected delivery date
attachment_urls	TEXT[]		YES	Array of valid URLs	Reference assets or brief documents
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of last update (UTC)

```
-- designer_task
CREATE INDEX idx_designer_task_assigned_to_user_id ON designer_task (assigned_to_user_id);
```

Table: designer_task_removal

Purpose: Audit log for designer task removals. Rather than hard-deleting tasks, removals are recorded here with a reason, preserving historical accountability.

Column Name	Data Type	Key	Null abl e	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique removal record
designer_task_id	UUID	FK	NO	FK → designer_task.id	The task that was removed
removed_by_user_id	UUID	FK	NO	FK → user.id	User who initiated the removal
removed_user_id	UUID	FK	NO	FK → user.id	User who created this record
reason	TEXT		NO	NOT NULL	Justification for removing the task
created_at	TIMESTAM PTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAM PTZ		NO	DEFAULT NOW()	Timestamp of last update (UTC)

Table: designer_application

Purpose: Designer applications for available tasks or projects. A designer may apply to be assigned work; CEO reviews and assigns accordingly.

Column Name	Data Type	Key	Null abl e	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique application record
designer_task_id	UUID	FK	NO	FK → designer_task.id	Task being applied for

Column Name	Data Type	Key	Null able	Constraints	Description
applicant_user_id	UUID	FK	NO	FK → user.id (role=designer)	Designer submitting the application
cover_note	TEXT		YES		Optional message from designer to CEO
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

```
CREATE INDEX idx_designer_application_task_applicant
ON designer_application (designer_task_id, applicant_user_id);
```

Table: designer_submission

Purpose: Work submissions by designers for their assigned tasks. Multiple submissions per task are allowed (e.g., revisions after feedback).

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique submission identifier
designer_task_id	UUID	FK	NO	FK → designer_task.id	The task this submission is for
stage	enum(case study , designing , rendering , final stage)		YES		The current stage that design task is
description	TEXT		NO	NOT NULL	Submission notes or change summary
attachment_urls	TEXT[]		YES	Array of valid URLs	Design files and deliverables

Column Name	Data Type	Key	Null able	Constraints	Description
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

```
CREATE INDEX idx_designer_submission_task_id
ON designer_submission (designer_task_id);
```

Table: designer_submission_review

Purpose: CEO review of a designer submission. Captures the review decision and feedback for each submission cycle.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique review record
designer_submission_id	UUID	FK	NO	FK → designer_submission.id	The designer submission being reviewed
reviewer_user_id	UUID	FK	NO	FK → user.id	CEO user performing the review
description	TEXT		NO	NOT NULL	Review feedback and notes
review_outcome	ENUM('approved','rejected','feedback','pending')		NO	NOT NULL	Decision for this submission cycle
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

Indexes:

```
CREATE INDEX idx_designer_submission_review_submission_id
ON designer_submission_review (designer_submission_id);
```

Table: designer_task_review

Purpose: Performance review of a designer's overall work on a completed task. Distinct from submission review — this evaluates the designer's quality, timeliness, and professionalism at task level.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique performance review record
designer_task_id	UUID	FK	NO	FK → designer_task.id	The completed task being evaluated
reviewer_user_id	UUID	FK	NO	FK → user.id	CEO reviewing performance
description	TEXT		yes	NULL	Performance evaluation narrative
Creativity	SMALLINT		no	CHECK (rating BETWEEN 1 AND 5)	Numeric performance score (1–5)
Timeliness	SMALLINT		no	CHECK (rating BETWEEN 1 AND 5)	Numeric performance score (1–5)
Rendering quality	SMALLINT		no	CHECK (rating BETWEEN 1 AND 5)	Numeric performance score (1–5)
Client understanding	SMALLINT		no	CHECK (rating BETWEEN 1 AND 5)	Numeric performance score (1–5)
review_outcome	SMALLINT		NO	Average of the 4 value	Overall performance verdict
created_at	TIMESTAMP PTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMP PTZ		NO	DEFAULT NOW()	Timestamp of last update (UTC)

Indexes:

CREATE INDEX idx_designer_task_review_task_id
ON designer_task_review (designer_task_id);

Table: paused_task

Purpose: Records task pause events for designer tasks. When a task is paused (e.g., waiting on materials or client decisions), the reason and timing are logged here for reporting and SLA tracking.

Column Name	Data Type	Key	Nullable	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique pause event record
designer_task_id	UUID	FK	NO	FK → designer_task.id	The task being paused
reason	TEXT		NO	NOT NULL	Reason the task was paused
paused_at	TIMESTAMP PTZ		NO	DEFAULT NOW()	Exact time the task was paused
resumed_at	TIMESTAMP PTZ		YES		Time the task was resumed (null if still paused)
created_at	TIMESTAMP PTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMP PTZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

2.6 CEO / Quantity Surveyor

Table: quantity_surveyor_task

Purpose: Tasks assigned to quantity surveyors by the CEO for site measurement, costing, or bill of quantities preparation.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique QS task identifier
assigned_to_user_id	UUID	FK	NO	FK → user.id (role=quantity_surveyor)	Quantity surveyor assigned
assigned_by_user_id	UUID	FK	NO	FK → user.id (role=ceo)	CEO who assigned the task
title	VARCHAR(500)		NO	NOT NULL	Short descriptive title
description	TEXT		NO	NOT NULL	Detailed scope of QS work required
status	ENUM('approved','rejected','feedback','pending')		yes	DEFAULT open	Task lifecycle status
task_state	Enum (active / deactive)				The current task is not rejected it is deactivated because of many reasons
due_date	DATE		no		Expected completion date
attachment_urls	TEXT[]		no	Array of valid URLs	Site plans or reference documents
created_at	TIMESTAMP PTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMP PTZ		yes	DEFAULT NOW()	Timestamp of last update (UTC)

Indexes:

```
CREATE INDEX idx_quantity_surveyor_task_assigned_to
ON quantity_surveyor_task (assigned_to_user_id);
```

Table: quantity_surveyor_submission

Purpose: Work submissions by quantity surveyors for their assigned tasks, such as BOQ documents, cost estimates, or measurement reports.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique submission record
quantity_surveyor_task_id	UUID	FK	NO	FK → quantity_surveyor_task.id	The QS task this submission belongs to
description	TEXT		NO	NOT NULL	Summary of submitted work
attachment_urls	TEXT[]		YES	Array of valid URLs	BOQ files, reports, spreadsheets
created_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAMPZ		NO	DEFAULT NOW()	Timestamp of last update (UTC)

Indexes:

```
CREATE INDEX idx_qs_submission_task_id
ON quantity_surveyor_submission (quantity_surveyor_task_id);
```

Table: quantity_surveyor_review

Purpose: CEO review of quantity surveyor submissions. Records decisions and feedback to guide revisions or mark completion.

Column Name	Data Type	Key	Null able	Constraints	Description
id	UUID	PK	NO	PRIMARY KEY	Unique review record
quantity_surveyor_submission_id	UUID	FK	NO	FK → quantity_surveyor_submission.id	The QS submission being reviewed
reviewer_user_id	UUID	FK	NO	FK → user.id (role=ceo)	CEO user reviewing the submission
description	TEXT		NO	NOT NULL	Review notes and feedback
review_outcome	ENUM('approved','rejected')		NO	NOT NULL	Outcome of the review

Column Name	Data Type	Key	Null abl e	Constraints	Description
	ted','feedback',pending)				
created_at	TIMESTAM PTZ		NO	DEFAULT NOW()	Timestamp of record creation (UTC)
updated_at	TIMESTAM PTZ		NO	DEFAULT NOW()	Timestamp of last update (UTC)

Indexes:
CREATE INDEX idx_quantity_surveyor_review_submission_id
ON quantity_surveyor_review (quantity_surveyor_submission_id);

Notification

Notification and Unread Activity Tracking Requirements

1. Notification Requirement

A notification system must be implemented to inform users whenever a task, submission, or review is created.

Notification Entity

Each notification record shall contain the following fields:

Field Name	Description
id	Unique notification identifier

user_id	User who should receive the notification
from_user_id	User who triggered the notification
resource_id	Related resource ID (task, submission, review, etc.)
resource_type	Type of resource (task, submission, review, customer, etc.)
notification_type	Notification category (Task Assigned, Submission Created, Review Added, Customer Created, etc.)
viewed	Indicates whether the notification has been viewed
created_at	Date and time the notification was created

Table: Notification model

```

CREATE TABLE notification (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL REFERENCES "user"(id),    -- recipient
  from_user_id UUID NOT NULL REFERENCES "user"(id),    -- actor (who triggered it)
  resource_id  UUID NOT NULL,                          -- submission_id, review_id, task_id
  resource_type VARCHAR(50) NOT= NULL,                  -- 'task_assigned' | 'submission' | 'review' | "posted job" | "apply" | 'paused' | 'rate'
  parent_id    UUID NOT NULL,                          -- always the ROOT task/card id
  parent_type  VARCHAR(50) NULL,                        -- 'data_collector_task' | 'designier_task' | 'qunatity_surveyor_task' | 'marketing_task'
  type        VARCHAR(255) NOT NULL,                    -- human-readable label
  viewed      BOOLEAN NOT NULL DEFAULT FALSE,
  created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Indexes for fast badge and item lookups
CREATE INDEX idx_notif_user_parent  ON notification(user_id, parent_id, viewed);
CREATE INDEX idx_notif_user_resource ON notification(user_id, resource_id, viewed);
CREATE INDEX idx_notif_user_resource_type ON notification(user_id, resource_type, viewed);
CREATE INDEX idx_notif_user_parent_type ON notification(user_id, parent_type, viewed);

```

